EXP NO:17 Experiment to understand Linear Regression for a given data set.

AIM:

To make an experiment to understand Linear Regression for a given data set.

PROGRAM:

```
from google.colab import files
uploaded = files.upload()
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
df.info()
df.dropna(inplace=True)
df.info()
df.describe()
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_stat
e=0)
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
model.score(x_train,y_train)
model.score(x_test,y_test)
model.coef_
model.intercept_
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_np=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_np)
print("Estimated Salary for {} years of experience is {}: ".format(yr_of_exp,Salary))
```

OUTPUT:

 Salary_data.csv(text/csv) - 362 bytes, last modified: 10/12/2026 - 100% done
Saving Salary_data.csv to Salary_data.csv

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
```

```
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
Enter Years of Experience: 4.4
Estimated Salary for 4.4 years of experience is [[67755.42970824]]:
```

RESULT:

Thus the experiment to understand Linear Regression for a given data set is done.